

High molecular weight DNA extraction for marine macroalgal tissue

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Marine macroalgae contain a variety of unique cell wall components including sulfated polysaccharides and polyphenolics. These components often co-elute with high molecular weight (HMW) DNA and lead to reduced library prep and sequencing outcomes. This protocol incorporates polyvinylpolypyrrolidone (PVPP) and β -mercaptoethanol (BME) to reduce polyphenolic contamination, and an early salting out step with potassium acetate (KOAc) to address polysaccharides.¹ This protocol is largely adapted from an Oxford Nanopore HMW DNA extraction from *Arabidopsis* leaves, which incorporates the QIAGEN Blood and Cell Culture DNA Midi Kit for column cleanup.² The DNA product often requires additional cleanup after elution, and we suggest the BluePippin 15kb size selection for all HMW applications.

Equipment:

- Lyophilizer
- Stir plate
- Heat block or water bath
- Vortex
- Mortar and pestle
- Refrigerated centrifuge for spins up to 3,500 x g with 50mL
- Suggested: Sage Science BluePippin

Consumables:

- Stock solution: 1M Tris-HCl, pH 9.5
- Stock solution: 5M sodium chloride (NaCl)
- Stock solution: 500mM ethylenediaminetetraacetic acid (EDTA)
- Stock solution: 5M potassium acetate (KOAc)
- Cetyltrimethylammonium bromide (CTAB)
- Polyethylene glycol (PEG) 8000
- β -mercaptoethanol (BME)
- Polyvinylpolypyrrolidone (PVPP)
- RNase A, 100mg/mL (eg. QIAGEN Mat. #1007885)
- 100% isopropanol
- 95-100% ethanol
- Nuclease-free water
- QIAGEN Blood and Cell Culture DNA Midi Kit (Cat. #13343)
- Tris-EDTA (TE) buffer
- 50 mL Falcon Tubes
- 1.5 mL collection tubes (eg. Eppendorf DNA LoBind tubes Cat. #022431021)
- Suggested: Sage Science High Pass Plus Cassette (BPLUS10 or BPLUS03) for BluePippin

Lyophilizing algal tissue:

1. Flash-freeze algal tissue in liquid nitrogen (target $\geq 5\text{g}$ wet tissue).
2. Quickly transfer sample to lyophilization container and freeze dry for 36-48 hours.
3. Macerate the tissue with a clean spatula to increase surface area and put on the lyophilizer for another 24 hours.
4. Remove and refrigerate with desiccant for immediate use, or store at -80°C for longer periods.

Setting up the DNA extraction:

1. Prepare desired volume of Carlson lysis buffer (100 mM Tris-HCl, pH 9.5, 2% CTAB, 1.4 M NaCl, 1% PEG 8000, 20 mM EDTA) and mix overnight on a magnetic stirrer. The stock solutions suggested under consumables will yield a homogenous buffer with no precipitate.
2. Pre-heat a heat block or water bath to 65°C and place in a fume hood.
3. For each extraction, transfer 20 ml of Carlson lysis buffer to a 50 ml Falcon tube.
4. In a fume hood, add 400 μl BME (originally 50uL) and mix by vortexing. Pre-warm the solution to 65°C for 30 minutes before starting the extraction.
5. Scoop 0.5 teaspoons lyophilized plant tissue into a clean mortar and add 50-100mg powdered PVPP. Grind with pestle for ~ 30 seconds, until tissue is powdered and combined, but not long enough to introduce significant moisture. Move immediately into DNA extraction.

Lysis and first precipitation:

1. Pour tissue into the warm buffer. Invert 5 times.
2. Add 40 μl of RNase A and vortex for 5 seconds.
3. Optional: If using a heat block with mixing, set the block (still at 65°C) to mixing at 300rpm for 5 minutes.
4. Incubate for 1 hour at 65°C . Invert 10 times every 15 minutes. At 30 minutes, add another 40 μL of RNase A, inverting 10 times to combine.
5. Allow the tubes to cool down to room temperature for 10 minutes.
6. Add 20 mL chloroform and vortex for two pulses of 5 seconds each.
7. Centrifuge the tubes at $3500 \times g$ for 15 minutes at 4°C .
8. In a fume hood, transfer the top layer of lysate from each tube to a new 50 ml Falcon tube, without disturbing the interphase.
Tip: The lysate layer should be 14-18mL of solution, but it is recommended to use wide-bore tips, transferring 1mL at a time. Tips can also be widened by cutting standard P1000 tips.
9. Mix supernatant with 0.4X 5M potassium acetate (KOAc) at room temperature, inverting at least 10 times to combine, then incubate on ice for 20 min.
10. Centrifuge the tubes at $3500 \times g$ for 45 minutes at 4°C .
11. Remove and retain the supernatant.
Tip: This may best be done by pouring slowly and observing the polysaccharide-salt pellet, which may be mobile. Leave some liquid behind in favor of avoiding the pellet.
12. Add 0.7X volumes of isopropanol. Invert 10 times. Incubate at -80°C for 15 minutes. Do not extend this incubation.
13. Centrifuge the sample at $3500 \times g$ for 45 minutes at 4°C .
Tip: If available, a fixed-angle centrifuge will create a pellet on the wall of the tube that has greater surface area for dissolution in step 15 (as compared to a conical pellet at the base of a falcon tube from a swinging bucket).
14. Discard the supernatant without disturbing the pellet. Use sterile wipes to absorb the liquid on the tube walls, being careful not to disturb the pellet.

15. To each pellet, add 10 mL G2 buffer, from the QIAGEN kit. Incubate at 50°C for 30-60 minutes, or until the pellet is dissolved. Swirl the pellet to mix but do not try to pipette or vortex.

Column cleanup:

1. Equilibrate a QIAGEN Genomic-tip 100/G column with 4 mL of Buffer QBT.
2. Pour the DNA in G2 buffer through the equilibrated column and allow it to flow through with just gravity.
3. Once all the lysate has passed through, wash the column with 8 mL of Buffer QC.
4. Repeat the wash with another 8 mL of Buffer QC.
5. Place the column over a clean 50 mL Falcon tube, and elute the genomic DNA with 5 mL of Buffer QF, pre-warmed to 55°C.
6. Allow the eluate to cool down to room temperature.
7. Add 3.5 mL of isopropanol to the eluted DNA and mix by inverting the tube 10 times.
8. Incubate the tube at -20°C for at least 3 hours, or overnight.

Final precipitation:

1. Centrifuge the tube at 3500 x g for 45 minutes at 4°C.
2. Discard the supernatant without disturbing the pellet.
3. Add 4 mL of ice-cold 70% ethanol to the pelleted DNA and invert the tube 10 times.
4. Centrifuge at 3500 x g for 10 minutes at 4°C.
Tip: If using a swinging bucket centrifuge the DNA will pellet at the base of the tube and be easy to locate and resuspend. If using a fixed angle, mark the side of the tube that faces outwards in order to locate the pellet for washes and elution.
5. Discard the supernatant without disturbing the pellet. Use sterile wipes to dry the tube walls, being careful not to disturb the pellet.
6. Resuspend the DNA in 100 µL of TE buffer and incubate at room temperature, typically overnight.
7. Transfer the DNA into a nuclease-free 1.5 mL tube (DNA LoBind tube preferred) using a wide-bore tip, and store at 4°C.
Tip: Often, waiting a further 48 hours before quantifying on Nanodrop and Qubit will allow the DNA to further relax and yield the most accurate results.
8. Carry samples forward to BluePippin size selection, if available. This gel separation will retain DNA fragments greater than 15 kb and discard any residual contamination still evident on a Nanodrop trace. For these benefits, expect 50-70% loss of DNA.

Additional tips:

- In the field or in lab, it is vital to scrape off all surface epiphytes and wash the sample in clean water before flash freezing to reduce contaminants common in the marine environment that confound genome assembly.
- Marine macroalgae are incredibly diverse in biochemical content, so individual seaweeds may require troubleshooting. Suggested alterations include varying input tissue type or quantity, increasing CTAB or BME percent, or adding a second chloroform separation.
- It may be necessary to carry out extractions of the same tissue in parallel to yield sufficient DNA, especially when large losses from BluePippin are expected. It is not suggested to combine multiple extractions onto the same column, as this may lead to overloading and a dirty sample. This protocol as written, paired with BluePippin, has produced sequencing-quality DNA for Nanopore from a red alga *Porteria hornemanii* and a brown alga *Macrocystis pyrifera*. For *P. hornemanii*, a single 20 mL extraction

produced sufficient DNA for sequencing, but for *M. pyrifera*, three parallel extractions of 20 mL were necessary.

Citations:

1. Chekan, J. R. *et al.* Scalable Biosynthesis of the Seaweed Neurochemical, Kainic Acid. *Angew Chem Int Ed Engl* **58**, 8454–8457 (2019).
2. Nanopore, Arabidopsis Leaf gDNA.
https://community.nanoporetech.com/extraction_method_groups/plant-leaf-gDNA